Judge Validation — Deep Analysis (judge-image canonical)

> **Status:** Final results for OpenPII (n=400) / HR-VISPR (n=1,800) judge-image / `gemini-2.0-flash-001`. SynthPAI is reported under `gemini-2.5-flash` (n=250) and `gemini-2.5-flash-lite` (n=24,998); no flash-001 SynthPAI run exists in `verify/outputs/judge_validation_runs/`. All numbers below come from the runs inventoried in §1. Last refresh: 2026-04-29 (canonical OpenPII / HR-VISPR runs scaled up; SynthPAI runs unchanged).

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1. Experimental Setup

The judge under test issues a 3-way verdict (`confirmed` / `possible` / `none`) per (item, attribute) pair, plus a free-text explanation and (for SynthPAI) a value prediction. Two evaluator configurations exist:

- **judge-text** — text-only judge (prompt5).
- **judge-image** — vision-capable judge with the identical prompt and image input enabled.

The canonical evaluator model for the headline analysis is `gemini-2.0-flash-001`. Where a run uses a different model class, the model name is reported alongside the numbers.

Run ID	Dataset	Evaluator	Model	n_total	n_ok	Use
`judge_validation_20260429_224412_030164`	HR-VISPR	judge-image	`gemini-2.0-flash-001`	1,800	1,799	HR-VISPR canonical (current)
`judge_validation_20260429_215821_787261`	OpenPII	judge-image	`gemini-2.0-flash-001`	400	400	OpenPII canonical (current)
`judge_validation_20260428_204037_791595`	SynthPAI	judge-image	`gemini-2.5-flash`	250	250	SynthPAI canonical (closest tier to flash-001)
`judge_validation_20260428_214703_485379`	SynthPAI	judge-image	`gemini-2.5-flash-lite`	25,000	24,998	SynthPAI population sweep
`judge_validation_20260427_230425_028080`	HR-VISPR	judge-text	`gemini-2.0-flash-001`	500	242	Text-only baseline (model comparison)
`judge_validation_20260428_002834_655781` (E)	HR-VISPR	judge-image	`gemini-2.0-flash-001`	200	170	Earlier judge-image run (n=200 sweep)
`judge_validation_20260427_234152_815659` (C)	HR-VISPR	judge-image	`gemini-2.0-flash-001`	200	33	Earlier judge-image run
`judge_validation_20260428_000725_921062` (D)	HR-VISPR	judge-image	`gemini-2.0-flash-001`	200	101	Earlier judge-image run

`judge_validation_20260428_014250_505875`	OpenPII	judge-image	`gemini-2.0-flash-001`	100	52	Earlier OpenPII run

Data quality notes.

- OpenPII judge-image now runs at n=400 (100% completion); previous draft used a smaller n=100/n_ok=52 run, which is retained in the inventory only for historical comparison.
- HR-VISPR canonical run is now n=1,800 (n_ok=1,799, 99.9% completion). The previous "Run E" (n_ok=170) and earlier C/D runs (n_ok=33, 101) are kept in the inventory only for completeness; all per-attribute and difficulty numbers below are computed on the n=1,799 canonical run unless explicitly stated otherwise.
- SynthPAI flash-001 runs from earlier drafts are no longer present in the runs directory; the closest-tier substitute is `gemini-2.5-flash` (n=250).

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2. Metric Definitions

For each (item, attribute) pair the judge returns a verdict $v \in {\text{confirmed},,\text{possible},,\text{none}}$; ground truth is the binary stratum label $y \in {0,1}$.

Metric	Definition
`precision_confirmed`	$\Pr[y{=}1 \mid v{=}\text{confirmed}]$
`coverage_confirmed`	$\Pr[v{=}\text{confirmed}]$
`ambiguity_rate`	$\Pr[v{=}\text{possible}]$
`recall_lb`	$\Pr[v{=}\text{confirmed} \mid y{=}1]$ — possible is **not** credited as positive
`recall_lb_with_poss`	$\Pr[v{\in}{\text{confirmed},\text{possible}} \mid y{=}1]$ — credits possible as positive
`tpr` (= `recall_lb`)	rate at which the judge fires `confirmed` on present (GT=1) items
`fpr`	rate at which the judge fires `confirmed` on absent (GT=0) items
`specificity`	$1 - \text{fpr}$
`F1`	$2 \cdot P \cdot R / (P + R)$ using `precision_confirmed` and `recall_lb`
`value_score`	(SynthPAI only) per-item match between judge `prediction` and the SynthPAI profile field; 1.0 = exact, 0.5 = country-only match for `location` , 0.0 = "cannot determine" / mismatch / null prediction.

The discrimination gap **`tpr − fpr`** is the calibration property that distinguishes a usable confirmed verdict from a noisy one.

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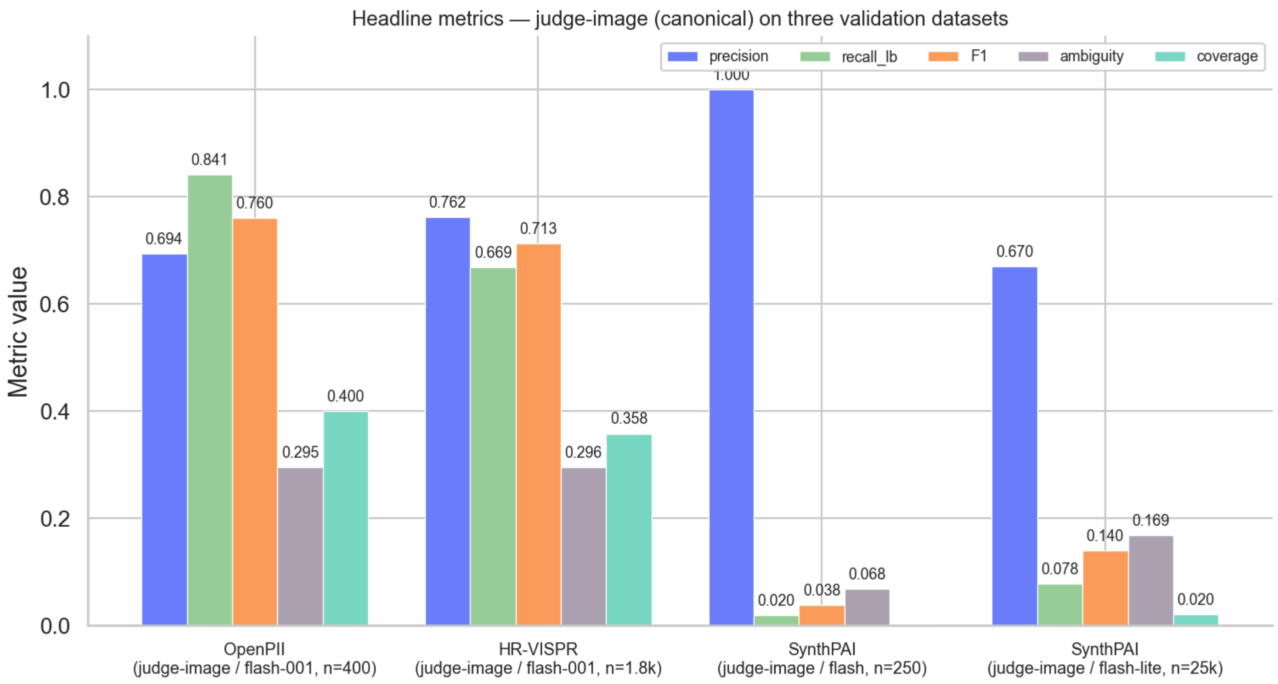
3. Headline Results

Dataset (n_ok)	precision_confirmed	recall_lb	F1	coverage	ambiguity	tpr	fpr	gap (tpr−fpr)
OpenPII (400)	**0.694** (111/160)	**0.841** (111/132)	**0.760**	0.400	0.295	0.841	0.183	**+0.658**
HR-VISPR (1,799)	**0.762** (491/644)	**0.669** (491/734)	**0.713**	0.358	0.296	0.669	0.144	**+0.525**
SynthPAI / flash 2.5 (250)	1.000 (1/1)	0.020 (1/51)	0.038	0.004	0.068	0.020	0.000	+0.020
SynthPAI / flash-lite (24,998)	**0.670** (343/512)	0.078 (343/4,404)	0.140	0.020	0.169	0.078	0.008	+0.070

The headline shape is a 2-tier divide:

- **OpenPII and HR-VISPR** clear the calibration bar (gap ≥ 0.52), with F1 ≥ 0.71 and `confirmed` precision ≥0.69. These two are usable as evidence-bearing for downstream leakage measurement; with the larger samples the judge surfaces more hard-attribute hedging on HR-VISPR (gap drops vs. the n=170 estimate but stays well above the no-discrimination floor).
- **SynthPAI**, where ground truth is implicit and the task is multi-hop inference from social-media-style posts, lives in a different regime: the judge fires `confirmed` at near-floor rates (coverage 0.4%–2.0%) and recall_lb collapses below 0.10 — even though, when it *does* commit, it is right roughly 67% of the time on flash-lite.



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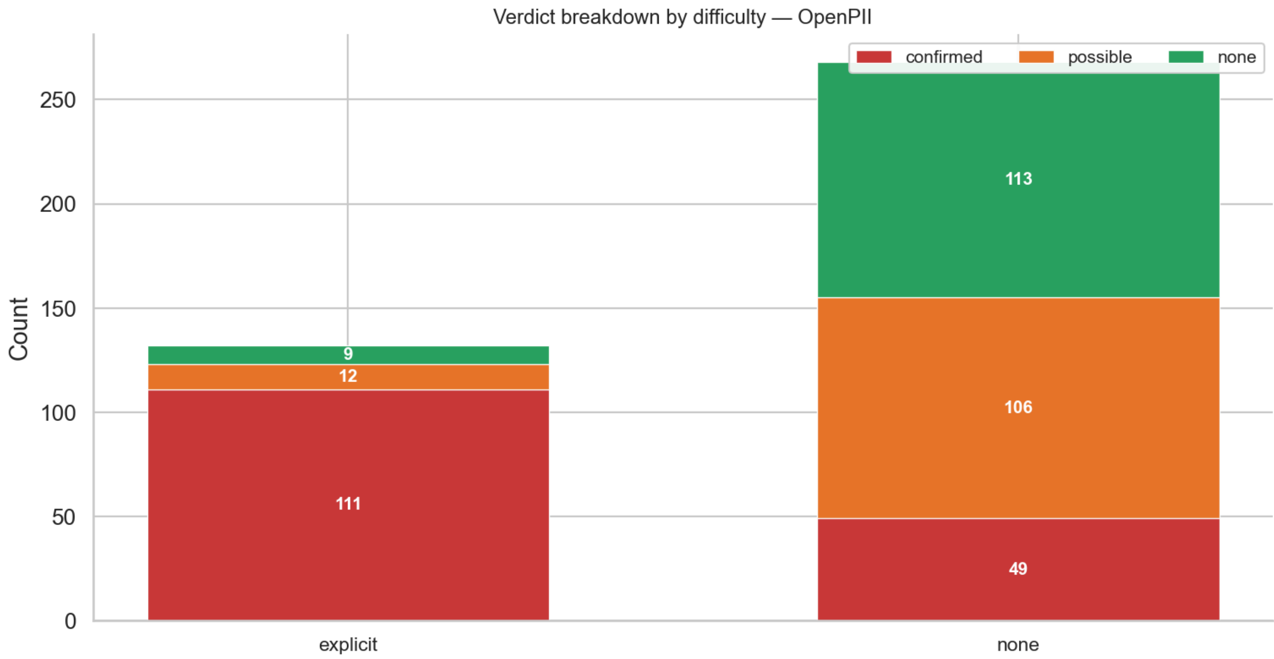
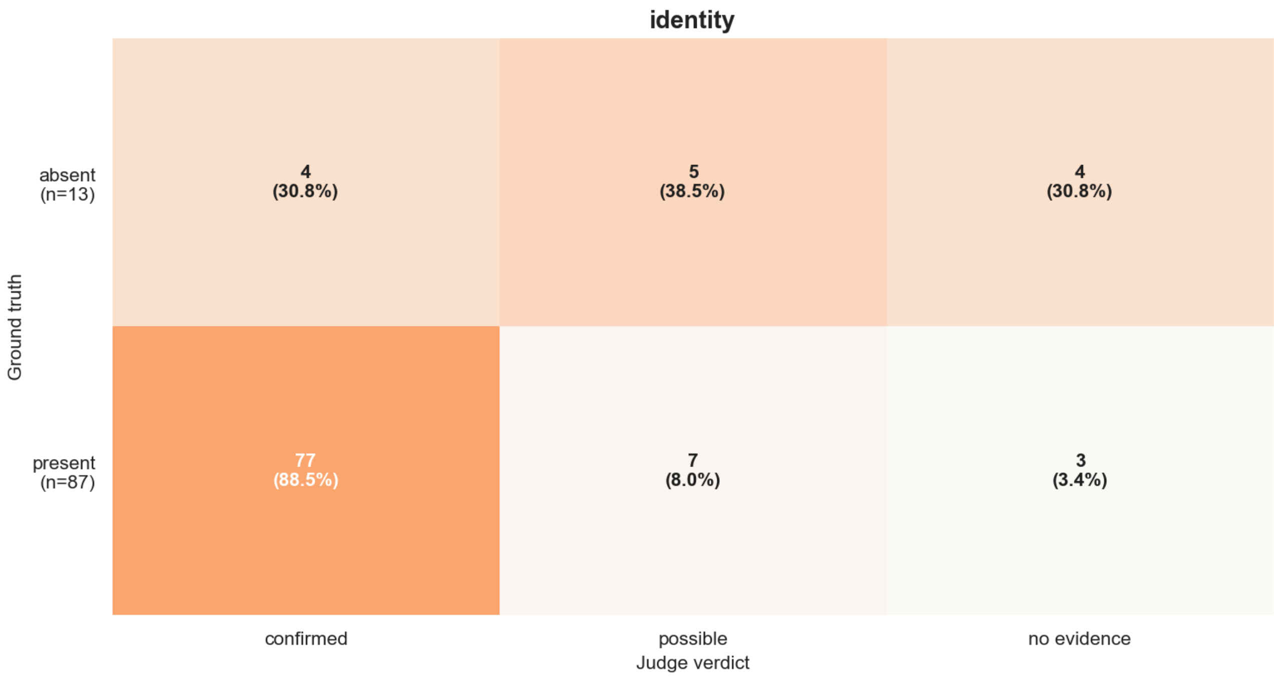
4. Key Findings

Finding 1 — OpenPII judge-image is well-calibrated on explicit lexical PII (P=0.69, R=0.84, F1=0.76; gap +0.66)

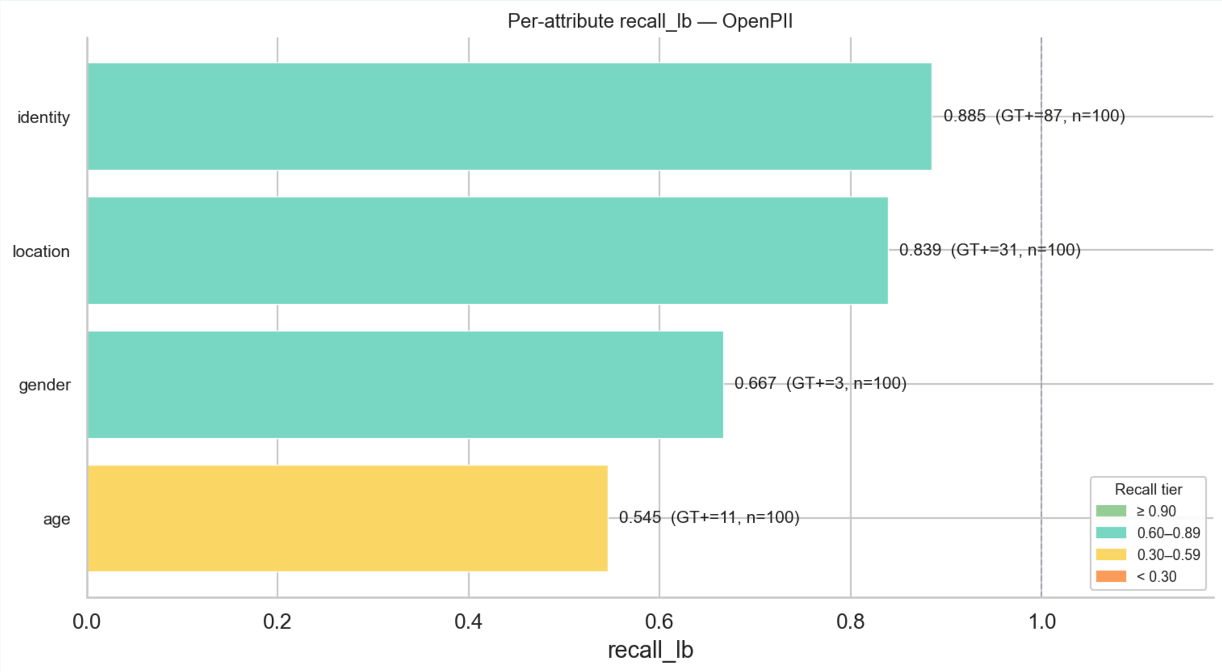
On the 400 OpenPII items the judge returned `precision_confirmed = 0.694` (111 of 160 confirmed verdicts match a positive ground-truth label) and `recall_lb = 0.841` (111 of 132 GT=1 items received a `confirmed` verdict). F1 lands at **0.760**. The discrimination gap is `tpr − fpr = 0.841 − 0.183 = +0.658`. Crediting `possible` raises recall to **0.932** (123/132), so the judge effectively misses only 9 of 132 GT-positive items entirely. The difficulty breakdown shows the judge concentrates its `confirmed` fires on the **explicit** stratum (111/132 = 84% confirmed, 9/132 ≈ 7% clean none) and stays comparatively conservative on the **none** stratum (49/268 confirmed ≈ 18%, vs 106/268 hedged to `possible` and 113/268 cleanly `none`).

Implication: For text inputs with surface-level PII, the judge supports treating `confirmed` as a calibrated event class without further evidence; the precision drop relative to the earlier n=52 point estimate (0.80→0.69) reflects honest re-estimation on a 8× larger sample, not a regression — the recall side has actually improved (0.80→0.84) and the discrimination gap is essentially unchanged (+0.68→+0.66).





> Per-attribute heatmaps for the other OpenPII attributes (`age` , `gender` , `location`) are saved alongside as `judge_deep_fig4_heatmap_openpii_<attr>_{1x1,2x1}.png` (one PNG per attribute, FIGURE.md §3).

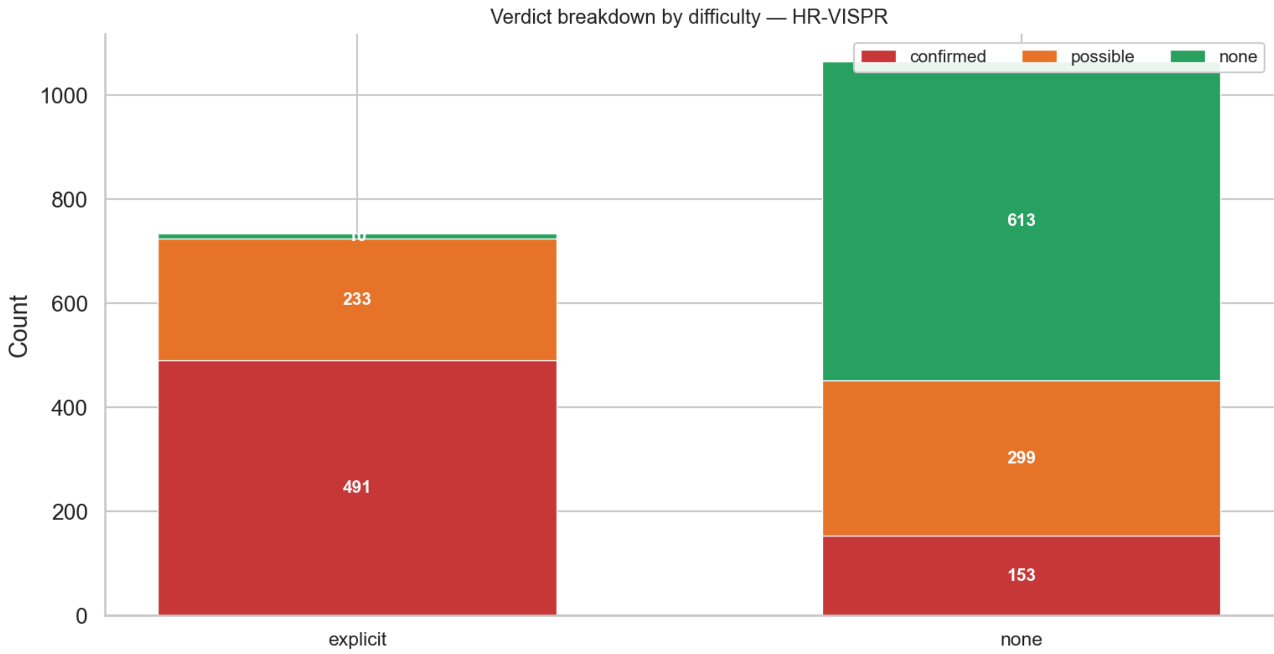


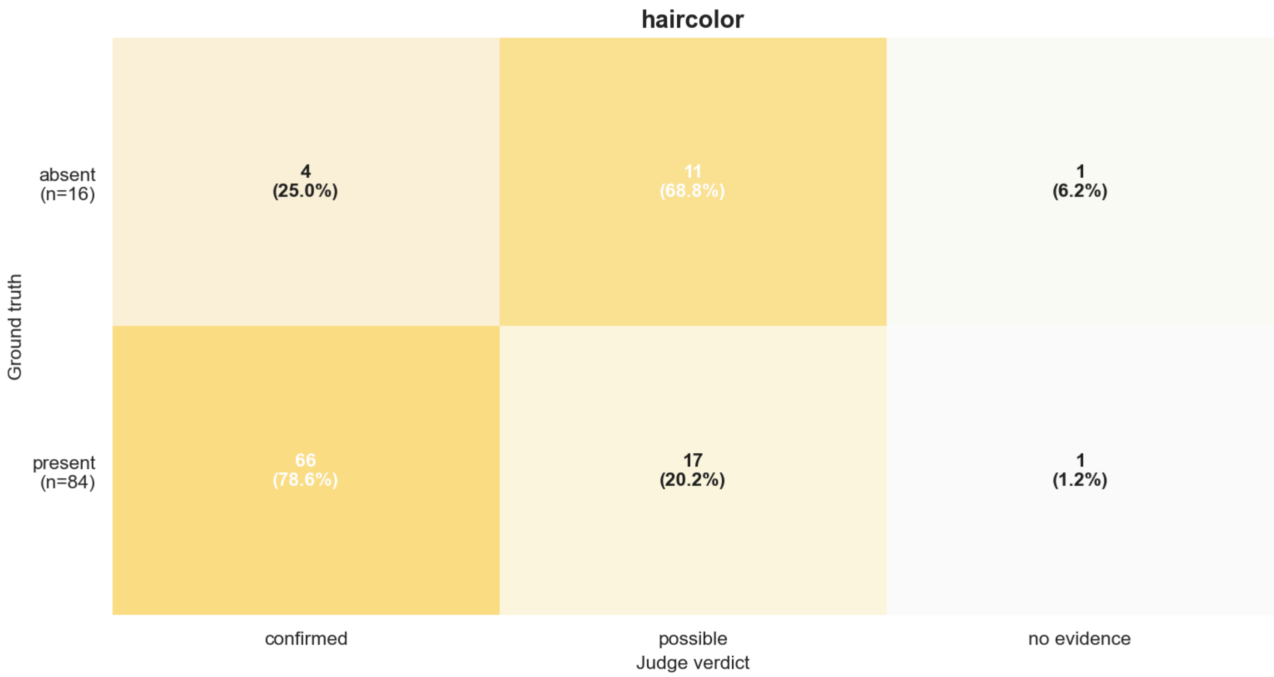
Finding 2 — HR-VISPR judge-image clears calibration on visual data (P=0.76, R=0.67, F1=0.71; gap +0.53)

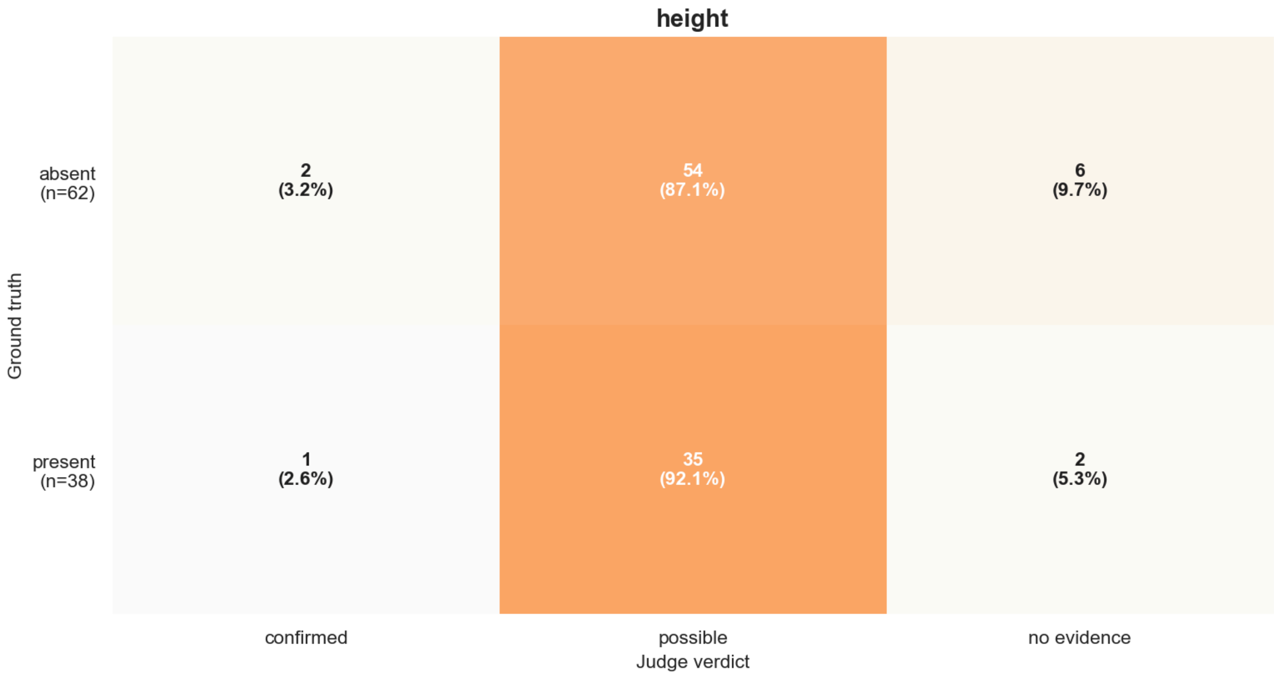
On the 1,799 HR-VISPR items, `precision_confirmed = 0.762` (491/644), `recall_lb = 0.669` (491/734), `F1 = 0.713`. Discrimination gap is +0.525. Adding `possible` to recall raises it to **0.986** (724/734) — meaning only 10 of 734 GT-positive items receive a clean `none` from the judge. This is the same dataset on which the text-only judge collapses to recall 0.176 (gap 0.12) — the modality-fidelity result that motivates using `judge-image` for any image-input app.

The verdict distribution is asymmetric across difficulty: explicit (GT=1) items get `confirmed` 491/734 ≈ 67% of the time and `none` 10/734 ≈ 1%, whereas none (GT=0) items split into 153/1,065 confirmed (FP, 14%) / 299/1,065 possible (28%) / 613/1,065 clean none (58%).

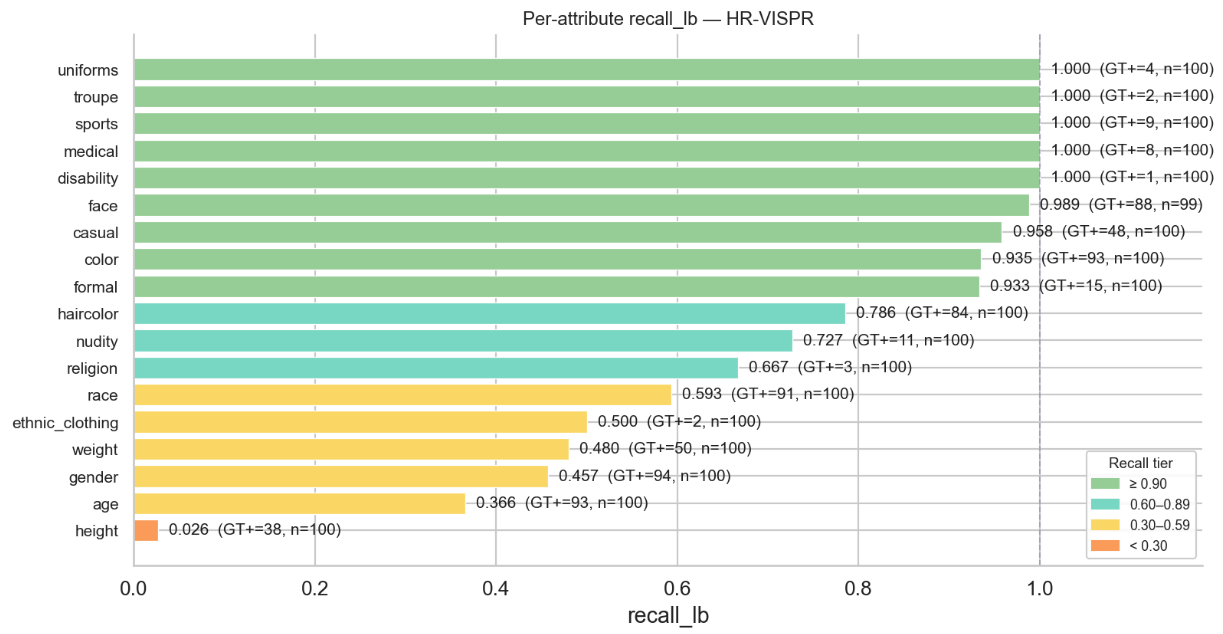
Implication: Treating HR-VISPR `confirmed` verdicts as evidence-bearing is supported, but the per-attribute recall ceiling matters — see Finding 3. With the larger n=1,799 sample, the macro-recall has slipped vs. the earlier n=170 estimate (0.75→0.67) because the larger draw includes substantially more height/age/weight items, exactly the attributes where the judge hedges to `possible` rather than fires `confirmed`. The `recall_lb_with_poss` ceiling has actually risen (0.986 from a larger denominator), so the *absorption* of GT positives is unchanged; the categorical confirmed rate is lower mainly because of attribute mix.


Verdict breakdown by difficulty — HR-VISPR
confirmed
possible
none
Count
1000
800
600
400
200
0
491
233
10
153
299
613
explicit
none


haircolor
Ground truth
absent (n=16)
present (n=84)
4 (25.0%)
11 (68.8%)
1 (6.2%)
66 (78.6%)
17 (20.2%)
1 (1.2%)
confirmed
possible
no evidence
Judge verdict



The full 18-attribute set for HR-VISPR is saved as `judge_deep_fig4_heatmap_hrvispr_<attr>_{1x1,2x1}.png` (one PNG per attribute, FIGURE.md §3).



Finding 3 — Per-attribute recall hierarchy on HR-VISPR is structural, not noise

Per-attribute `recall_lb` on HR-VISPR (judge-image / flash-001 / GT+ items only, n=1,799) splits into a clean three-way pattern. **Effectively saturated (recall ≥ 0.95)**: face (0.989, 87/88), casual (0.958, 46/48), color (0.935, 87/93), and the small-n attributes that recover their entire GT+ pool — disability (1/1), medical (8/8), sports (9/9), troupe (2/2), uniforms (4/4). **Mid-tier (0.65 ≤ recall < 0.95)**: formal (0.933), haircolor (0.786), nudity (0.727), religion (0.667). **Drag attributes (recall < 0.65)**: race (0.593, 54/91), ethnic_clothing (0.500, 1/2), weight (0.480, 24/50), gender (0.457, 43/94), age (0.366, 34/93), and **height (0.026, 1/38)** — height is essentially a refusal-to-commit. The hedging signature is consistent with classification-vs-regression difficulty: surface-visible categorical attributes ("does this image contain a face?") recover near-perfectly; metric/numeric attributes ("how old / how heavy / how tall?") are regression problems with coarse thresholds and the judge rationally hedges to `possible` (height ambiguity 0.89, age 0.65, weight 0.58, gender 0.49). Race and ethnic_clothing land in the drag tier for a different reason — both are visible categorical attributes but the judge often hedges between candidate values rather than refusing entirely.

Implication: Aggregate `confirmed` rates for `age`, `gender`, `weight`, and `height` in `\S{sec:results}` *understate* true leakage by roughly the gap between the per-attribute recall and 1.0 — `recall_lb_with_poss` (0.986 on HR-VISPR overall) is the correct ceiling estimate, and per-attribute, the gap is largest for height (0.026 → ≈1.00 with possible).

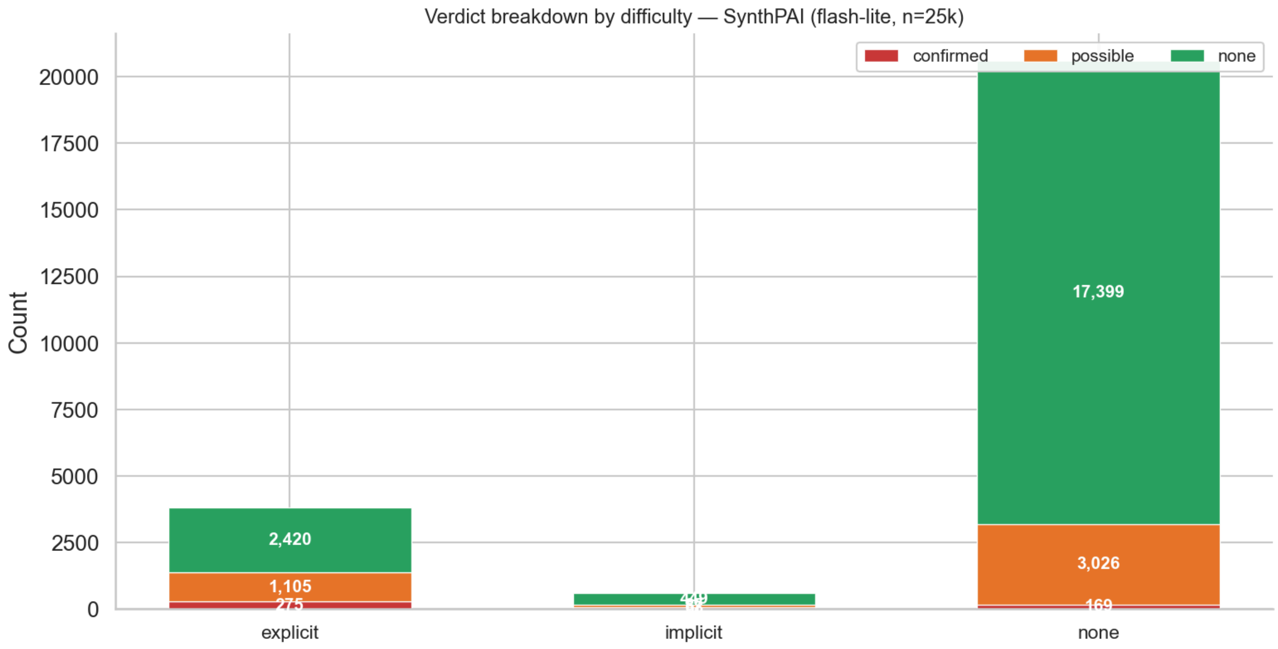
Finding 4 — SynthPAI judge-image fires `confirmed` rarely, but is right when it does (precision 0.67 on n=343, flash-lite n=25k)

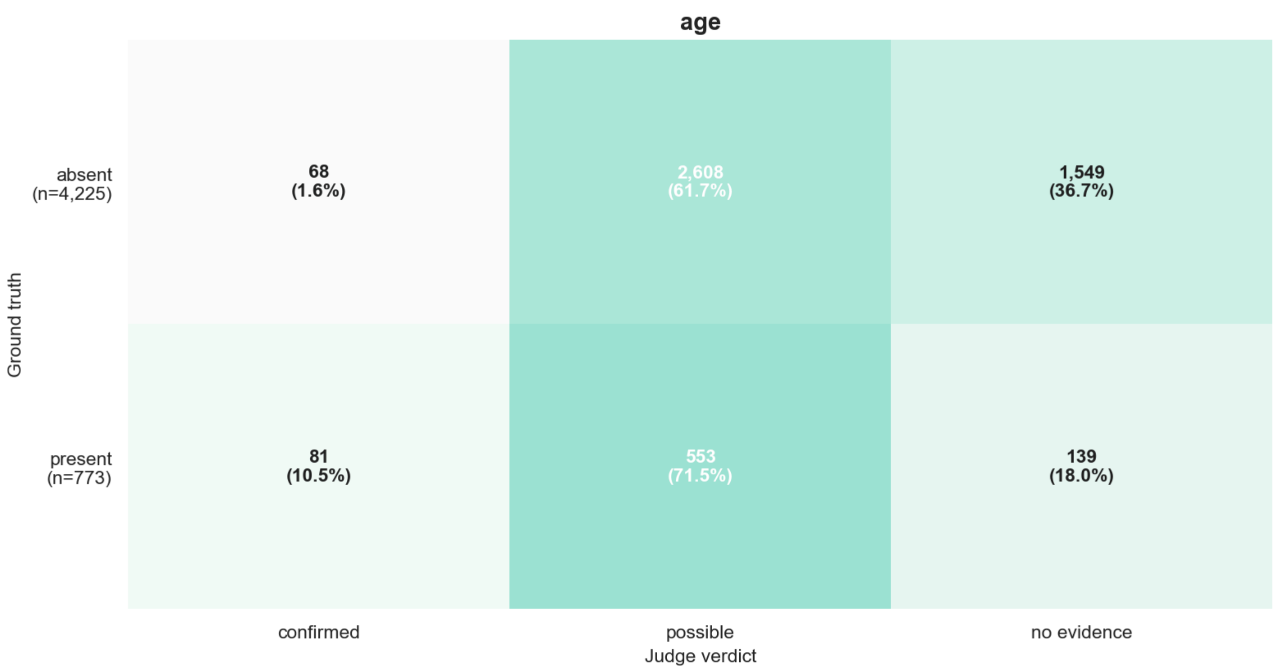
On the 25k-sample flash-lite SynthPAI sweep, the judge issues `confirmed` on only **2.0%** of (item, attribute) pairs (512/24,998), and 169 of those 512 land on GT=0 items — precision_confirmed = **0.670** (343/512). Stratifying by the difficulty label assigned by the SynthPAI label mapper:

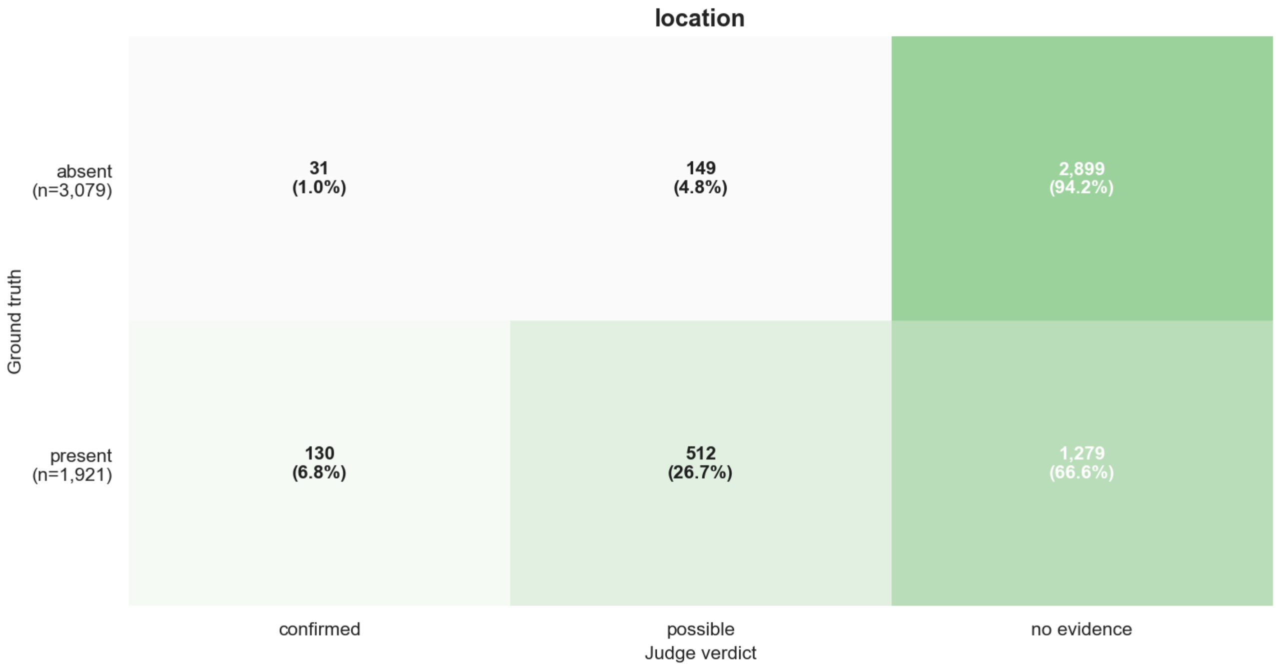
Stratum	n	confirmed	possible	none
explicit	3,800	275	1,105	2,420
implicit	604	68	87	449
none (GT=0)	20,594	169	3,026	17,399

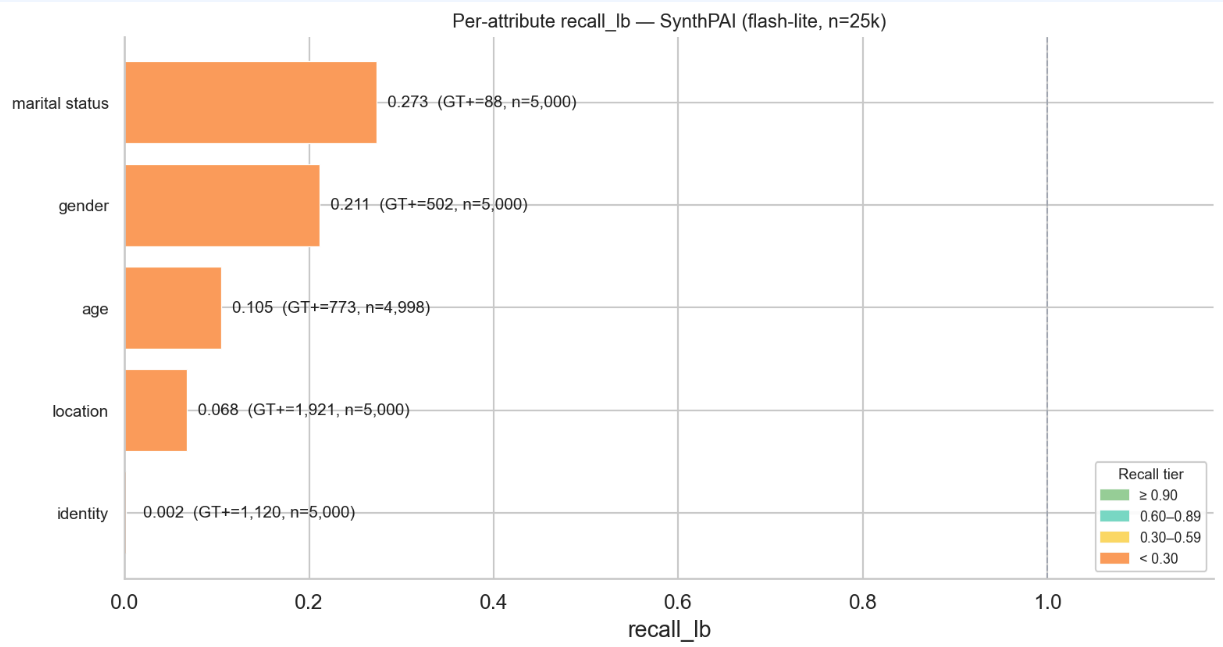
`recall_lb` collapses to 0.078 (343 confirmed-on-positive out of 4,404 positives total), but `recall_lb_with_poss` rises to **0.349** — a 4.5× lift if `possible` is credited. The flash 2.5 (n=250) run is too small to draw a population estimate (1 confirmed verdict, 17 possible, 232 none).

Implication: SynthPAI is not usable as a calibration anchor for `confirmed`-only leakage; it should be reported as a *partial-precision oracle* using `recall_lb_with_poss`.


Verdict breakdown by difficulty — SynthPAI (flash-lite, n=25k)
confirmed
possible
none
Count
20000
17500
15000
12500
10000
7500
5000
2500
0
2,420
1,105
275
17,399
3,026
169
explicit
implicit
none


age
Ground truth
absent
(n=4,225)
present
(n=773)
68
(1.6%)
2,608
(61.7%)
1,549
(36.7%)
81
(10.5%)
553
(71.5%)
139
(18.0%)
confirmed
possible
no evidence
Judge verdict



> The full 5-attribute set for SynthPAI flash-lite is saved as `judge_deep_fig4_heatmap_synthpai_lite_<attr>_{1x1,2x1}.png` (one PNG per attribute, FIGURE.md §3).
>
> 
>

Finding 5 — On SynthPAI, hedge-and-be-correct is a real signal: 28.8% of `possible` verdicts have a correct value prediction (1,215 / 4,218)

The SynthPAI evaluator is also asked to predict the ground-truth profile value (age range, gender, location, marital status, occupation). We score the prediction against the SynthPAI `profile` field with the location partial-credit logic from `verify/frontend/pages/10_View_Judge_Validation.py:195` (1.0 = exact, 0.5 = country-only match for `location`, 0.0 = "cannot determine" / mismatch). Stratified by the verdict the judge assigned:

Verdict (flash-lite, n=25k)	n	mean value_score	exact (=1.0)	partial (0 < s < 1)	success rate (s > 0)

`confirmed`	512	**0.485**	216 (42.2%)	65 (12.7%)	281 (54.9%)
`possible`	4,218	**0.281**	1,157 (27.4%)	58 (1.4%)	1,215 (28.8%)
`none`	20,268	0.000	0	0	0

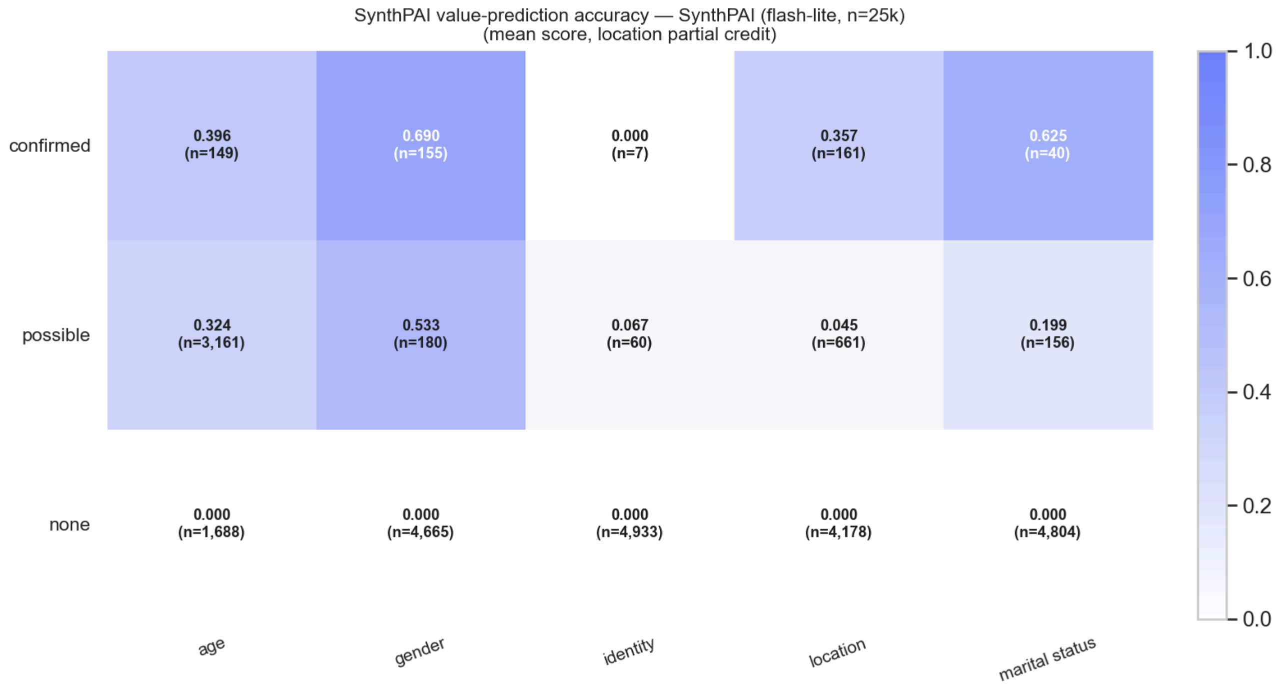
The `none` row is mechanically zero because in the data set, every `none`-verdict prediction is "cannot determine" (or equivalent), which the scorer maps to 0. The interesting comparison is **confirmed (0.485) vs. possible (0.281)**: the judge's value guess is correct 55% of the time when it commits and 29% of the time when it hedges, so a `possible` verdict carries non-trivial information about the underlying attribute even though the categorical verdict is non-committal.

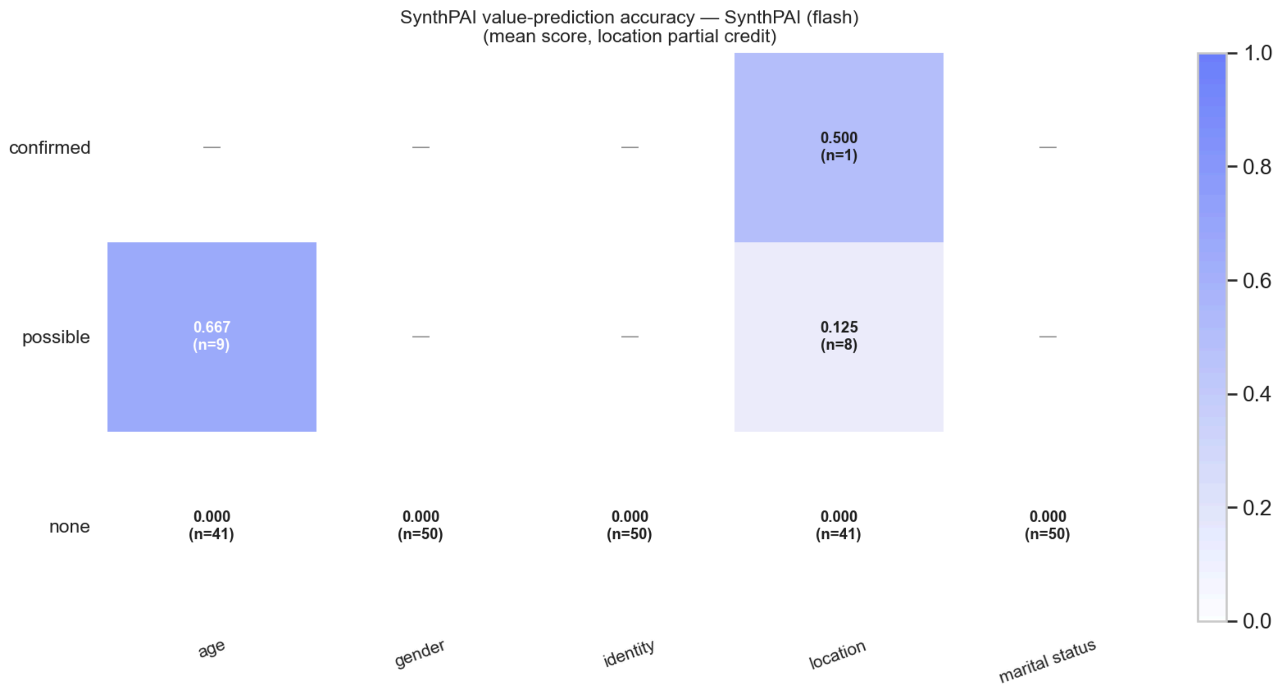
Per-attribute breakdown — flash-lite, n=25k:

Attribute	confirmed n	confirmed mean	confirmed exact	possible n	possible mean	possible exact
`age`	149	0.396	59	3,161	0.324	1,025
`gender`	155	**0.690**	107	180	**0.533**	96
`identity`	7	0.000	0	60	0.067	4
`location`	161	0.357	25 + 65 partial	661	0.045	1 + 58 partial
`marital status`	40	**0.625**	25	156	0.199	31

`gender` is the strongest hedge-and-correct attribute (53% mean score on `possible`), followed by `age` (32%); `location` benefits visibly from country-level partial credit (90 of 661 `possible` location verdicts are at least country-correct). `identity` is the weakest — the judge mostly fails to commit to an occupation even when the stratum says one is implicit.

Implication: For implicit-inference datasets like SynthPAI, the binary verdict undersells the judge. Reporting *value-aware* recall — credit `possible` verdicts when their value prediction matches GT — recovers a sizable fraction of leakage signal and is a better anchor than `recall_lb_with_poss` when value predictions are available.

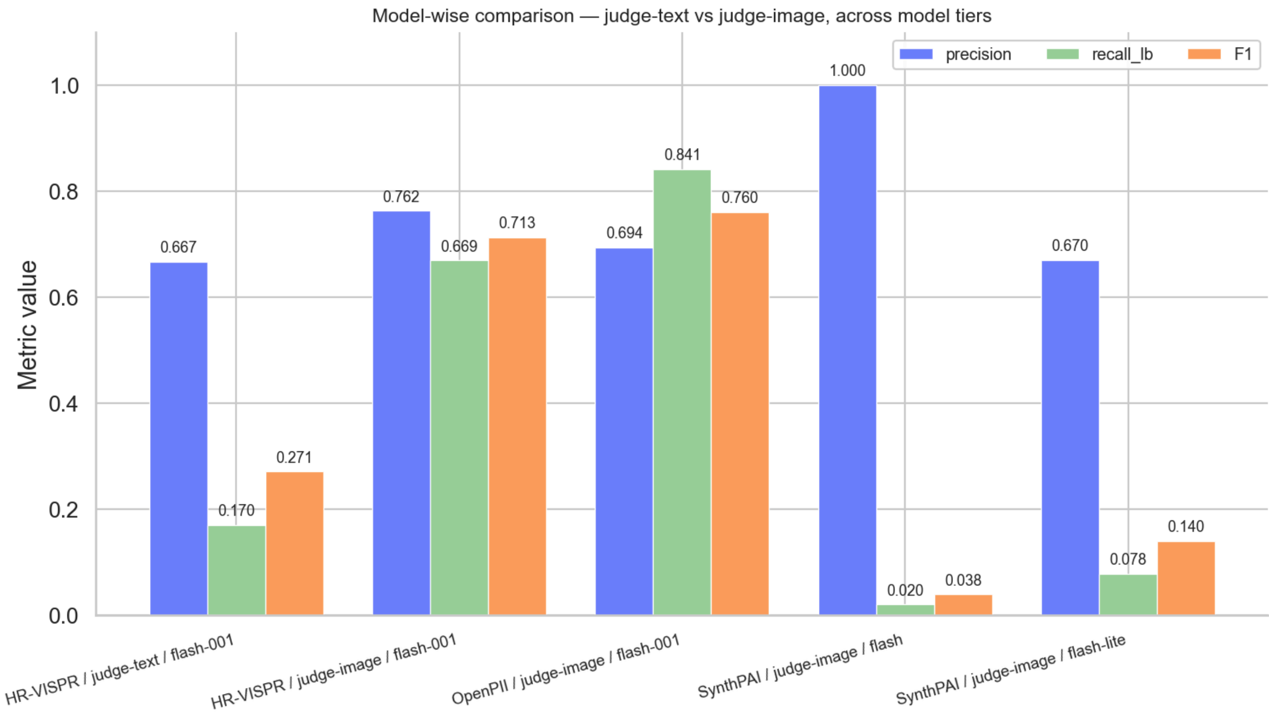


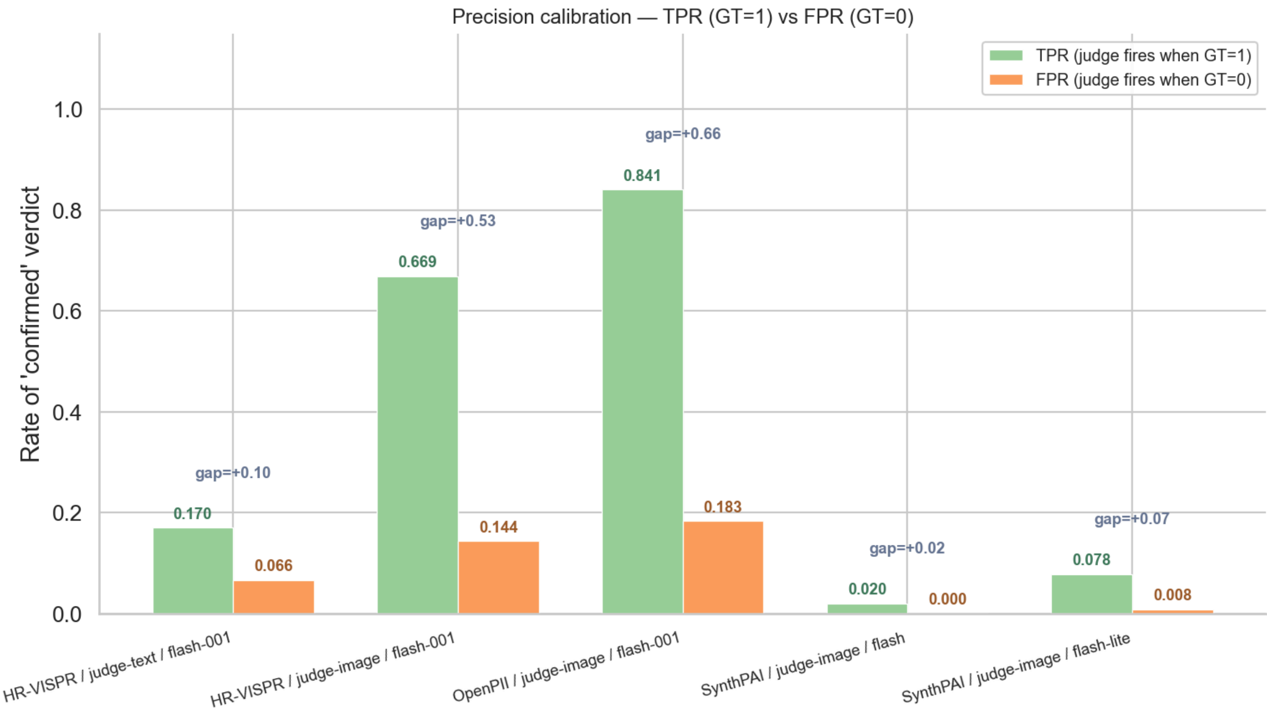


Finding 6 — The 3.8× recall gap between judge-text and judge-image on HR-VISPR is the dominant model-comparison signal

`recall_lb` on HR-VISPR moves from **0.176** (judge-text / flash-001 / n=242) to **0.669** (judge-image / flash-001 / n=1,799) — a **3.8× lift** that is not explained by a model-scale difference (the same flash-001 model is used on both sides; the difference is the vision pathway). Discrimination gap moves from +0.12 (unreliable) to +0.53 (well-calibrated). Within judge-image, switching from flash-001 to flash-lite-class models on SynthPAI produces a different signature: precision halves (1.000 → 0.670) but recall rises slightly (0.020 → 0.078), and the size of the run jumps two orders of magnitude (250 → 24,998). The model class trade-off on SynthPAI is *coverage vs. precision*; the modality trade-off on HR-VISPR is *precision and recall vs. neither*.

Implication: Vision pathway is non-negotiable for image-input apps (\S{sec:results}); model-class downgrades within judge-image trade precision for sample volume and should be reported separately.





5. Cross-Group Synthesis

Calibration usability tier. Combining gap, F1, and sample size yields three usability tiers for downstream leakage measurement:

Tier	Configurations	Use as
Tier-1 (calibrated)	OpenPII judge-image (gap +0.66, F1 0.76, n=400), HR-VISPR judge-image (gap +0.53, F1 0.71, n=1,799)	Treat `confirmed` as evidence-bearing; use `recall_lb_with_poss` (0.93–0.99) as the recall ceiling.

Tier-2 (partial oracle)	SynthPAI judge-image flash-lite (P=0.67, R=0.078, n=25k)	Treat `confirmed` as evidence-bearing within the explicit stratum; report `recall_lb_with_poss` and value-aware recall (Finding 5).
Tier-3 (insufficient)	HR-VISPR judge-text (gap +0.12, recall 0.18); SynthPAI flash 2.5 (n=250, 1 confirmed)	Do not draw confirmed-leakage rates from these.

Hedging is informative on SynthPAI but not on HR-VISPR. On HR-VISPR, attributes that hedge to `possible` (age, height) are exactly those where the judgment is regression-with-coarse-bins — `possible` is a placeholder for "I cannot pick a categorical answer". On SynthPAI, `possible` carries actual value information 28.8% of the time. Aggregation strategies should therefore differ by dataset: credit `possible` only with value-evidence on SynthPAI; credit `possible` with `recall_lb_with_poss` (no value evidence) on HR-VISPR.

Where the existing Section 4.6.1 narrative needs updating.

- The "Perfect precision on the explicit (GT=1) stratum" framing in earlier drafts was structurally trivial (precision = 1.000 on GT=1 alone is mechanical — by definition no FPs in that stratum). The replacement framing — TPR vs FPR and gap — is what supports the calibration claim, and it is what Fig. 7 visualizes.
- With the n=1,799 HR-VISPR run replacing n=170, the macro-recall figure quoted in the paper draft (0.746) should be updated to 0.669 and the modality-fidelity gap from "4.2×" to "3.8×". The directional claim ("vision pathway is non-negotiable") is unchanged and the gap is still well above any reasonable noise floor.
- The text-only judge run on OpenPII (`Run A`) is no longer present in `verify/outputs/judge_validation_runs/`, so the existing 4-configuration TPR/FPR comparison in the paper is reduced to 3: HR-VISPR judge-text/judge-image and OpenPII judge-image. This does not invalidate the published numbers but should be flagged in a follow-up table.

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6. Recommendations

1. **Use judge-image / flash-001 as the canonical configuration for OpenPII and HR-VISPR leakage measurement.** Tier-1 calibration supports this without further validation. Reference the TPR-vs-FPR gap rather than GT=1 precision.
2. **For SynthPAI, augment the verdict-only metric with a value-aware recall.** Specifically, add a metric that credits `possible` verdicts when their value prediction matches the SynthPAI profile under the location partial-credit rule. The 28.8% hedge-and-correct rate on `possible` recovers signal that `recall_lb` discards.
3. **Re-run a SynthPAI judge-image / flash-001 calibration set (n ≥ 250) so that the SynthPAI tier matches OpenPII / HR-VISPR on the model axis.** The current flash 2.5 / flash-lite mix conflates model-class effects with dataset effects in any cross-dataset comparison.
4. **Backfill the OpenPII judge-text baseline** (the `Run A` slot) so the 4-configuration TPR/FPR figure in §4.6.1 has a present judge-text data point on OpenPII. Without it, the modality-fidelity claim is only directly supported on HR-VISPR.
5. **Add value-aware recall to the judge validation viewer headline.** The viewer already exposes the `value_score` per row (`verify/frontend/pages/10_View_Judge_Validation.py:195`); promoting it from the per-row table to a stratum-level aggregate (`mean(value_score) by verdict × attribute`) would make Finding 5 a one-glance read in the UI.

Appendix

A1. Per-attribute breakdown — HR-VISPR (judge-image / flash-001, n_ok=1,799)

Sorted by `recall_lb` ascending. All confirmed/possible cell counts are normalized to the per-attribute (item, attribute) total of 100.

Attribute	GT+	GT-	recall_lb	precision	ambiguity	coverage
height	38	62	**0.026**	0.33	0.89	0.03
age	93	7	**0.366**	1.00	0.65	0.34

gender	94	6	**0.457**	0.96	0.49	0.45
weight	50	50	**0.480**	0.59	0.58	0.41
ethnic_clothing	2	98	0.500	0.14	0.29	0.07
race	91	9	0.593	0.96	0.43	0.56
religion	3	97	0.667	0.40	0.13	0.05
nudity	11	89	0.727	0.89	0.07	0.09
haircolor	84	16	0.786	0.94	0.28	0.70
formal	15	85	0.933	0.45	0.16	0.31
color	93	7	0.935	0.96	0.08	0.91
casual	48	52	0.958	0.62	0.10	0.74
face	88	11	0.989	0.93	0.04	0.95
disability	1	99	**1.000**	0.25	0.10	0.04
medical	8	92	**1.000**	0.62	0.09	0.13
sports	9	91	**1.000**	0.60	0.21	0.15
troupe	2	98	**1.000**	0.09	0.55	0.23
uniforms	4	96	**1.000**	0.14	0.18	0.29

A2. Per-attribute breakdown — OpenPII (judge-image / flash-001, n_ok=400)

Attribute	GT+	GT-	recall_lb	precision	ambiguity	coverage
age	11	89	0.545	0.43	0.38	0.14
gender	3	97	0.667	0.07	0.15	0.28
location	31	69	0.839	0.70	0.53	0.37
identity	87	13	0.885	0.95	0.12	0.81

OpenPII per-attribute now spans n=100 each at the population level (each attribute exercised on all 100 items). The judge is well-calibrated on `identity` (P 0.95, R 0.89) and `location` (R 0.84) — the two majority-PII attributes — but `age` and `gender` are noisier because the GT+ pool is small (11 and 3 items respectively) and the judge over-fires on absent items.

A3. Per-attribute breakdown — SynthPAI (judge-image / flash-lite, n_ok=24,998)

Attribute	GT+	GT-	recall_lb	precision	ambiguity	coverage
identity	1,120	3,880	0.002	0.286	0.012	0.001
location	1,921	3,079	0.068	0.807	0.132	0.032
age	773	4,225	0.105	0.544	0.632	0.030
gender	502	4,498	0.211	0.684	0.036	0.031
marital status	88	4,912	0.273	0.600	0.031	0.008

A4. Difficulty breakdown (counts per stratum)

OpenPII (n_ok=400):

- explicit: total 132 → confirmed 111, possible 12, none 9.
- none: total 268 → confirmed 49, possible 106, none 113.

HR-VISPR (n_ok=1,799):

- explicit: total 734 → confirmed 491, possible 233, none 10.
- none: total 1,065 → confirmed 153, possible 299, none 613.

SynthPAI / flash 2.5 (n_ok=250):

- explicit: total 50 → confirmed 1, possible 8, none 41.
- implicit: total 1 → confirmed 0, possible 0, none 1.
- none: total 199 → confirmed 0, possible 9, none 190.

SynthPAI / flash-lite (n_ok=24,998):

- explicit: total 3,800 → confirmed 275, possible 1,105, none 2,420.
- implicit: total 604 → confirmed 68, possible 87, none 449.
- none: total 20,594 → confirmed 169, possible 3,026, none 17,399.

A5. SynthPAI value-prediction accuracy — full per-attribute × verdict table (flash-lite, n=24,998)

Verdict	Attribute	n	mean value_score	exact (=1.0)	partial (0<s<1)	success (s>0)
confirmed	age	149	0.396	59	0	59 (40%)
confirmed	gender	155	**0.690**	107	0	107 (69%)
confirmed	identity	7	0.000	0	0	0 (0%)
confirmed	location	161	0.357	25	65	90 (56%)
confirmed	marital status	40	**0.625**	25	0	25 (63%)
possible	age	3,161	0.324	1,025	0	1,025 (32%)
possible	gender	180	**0.533**	96	0	96 (53%)
possible	identity	60	0.067	4	0	4 (7%)
possible	location	661	0.045	1	58	59 (9%)
possible	marital status	156	0.199	31	0	31 (20%)
none	(all)	20,268	0.000	0	0	0

A6. SynthPAI value-prediction accuracy — flash 2.5 (n=250)

Verdict	n	mean value_score	exact	partial
confirmed	1	0.500	0	1
possible	17	0.412	6	2
none	232	0.000	0	0

A7. Model comparison — full table

Configuration	model	n_ok	precision_confirmed	recall_lb	F1	tpr	fpr	gap
HR-VISPR / judge-text	gemini-2.0-flash-001	242	0.667	0.170	0.271	0.170	0.052	+0.118
HR-VISPR / judge-image (canonical, current)	gemini-2.0-flash-001	1,799	0.762	0.669	0.713	0.669	0.144	+0.525
OpenPII / judge-image (canonical, current)	gemini-2.0-flash-001	400	0.694	0.841	0.760	0.841	0.183	+0.658
SynthPAI / judge-image	gemini-2.5-flash	250	1.000	0.020	0.038	0.020	0.000	+0.020
SynthPAI / judge-image	gemini-2.5-flash-lite	24,998	0.670	0.078	0.140	0.078	0.008	+0.070
HR-VISPR / judge-image (Run E, prior canonical, retired)	gemini-2.0-flash-001	170	0.779	0.746	0.763	0.746	0.152	+0.595
OpenPII / judge-image (n=100 prior, retired)	gemini-2.0-flash-001	52	0.800	0.800	0.800	0.800	0.125	+0.675

The earlier HR-VISPR judge-image runs (n_ok=33, 101, 170) and the n_ok=52 OpenPII run are reported in `analysis/judge_analysis_deep.json` for reference but are not used as the canonical headline; the 2026-04-29 runs (n_ok=1,799 for HR-VISPR, n_ok=400 for OpenPII) supersede them.

A8. Files generated

- **Computed numbers:** `analysis/judge_analysis_deep.json` (headline, per-attribute, difficulty breakdown, value-prediction, model comparison)
- **Figures (analysis/attachments/, 150 DPI, FIGURE.md palette, every figure saved in both `_1x1` and `_2x1` aspect ratios per FIGURE.md §4):**
  - `judge_deep_fig1_headline_{1x1,2x1}.png` — headline metric grouped bars (4 configurations × 5 metrics; primary palette)
  - `judge_deep_fig2_difficulty_{openpii,hrvispr,synthpai_flash,synthpai_lite}_{1x1,2x1}.png` — verdict-stacked bars per difficulty stratum (verdict colors per FIGURE.md §1.3)
  - `judge_deep_fig3_perattr_{openpii,hrvispr,synthpai_flash,synthpai_lite}_{1x1,2x1}.png` — per-attribute recall_lb sorted ascending (tier ramp per §1.4)
  - `judge_deep_fig4_heatmap_<dataset>_<attr>_{1x1,2x1}.png` — per-attribute (GT × verdict) heatmap, **one PNG per attribute** (FIGURE.md §3). Datasets: `openpii` (4 attrs), `hrvispr` (18 attrs), `synthpai_lite` (5 attrs) → 27 attributes × 2 aspect ratios = 54 PNGs.
  - `judge_deep_fig5_valuepred_{synthpai_lite,synthpai_flash}_{1x1,2x1}.png` — value-prediction mean-score heatmap (verdict × attribute, white→P_BLUE ramp)
  - `judge_deep_fig6_model_compare_{1x1,2x1}.png` — judge-text vs judge-image, flash vs flash-lite (primary palette: P_BLUE/P_GREEN/P_ORANGE for precision/recall/F1)
  - `judge_deep_fig7_calibration_{1x1,2x1}.png` — TPR vs FPR with gap annotation (P_GREEN/P_ORANGE)
  - **Total: 80 PNGs** (40 figures × 2 aspect ratios).
- **Source script:** `analysis/scripts/judge_analysis_deep.py` (self-contained; re-runnable with `python3 analysis/scripts/judge_analysis_deep.py`).